



For further details, please check out:
www.semiconductors.philips.com/markets/identification
or e-mail info.bi@philips.com

ISO 14443

ISO 15693

Philips' Contactless Reader Systems

Product Features		Reader ICs		MF RC531		SL RC400		CL RC632		Evaluation Systems		MF EV700		MF EV800		SL EV400		SL EV900	
Operating distance typ.		100 mm		100 mm		100 mm		100 mm		75mm		75mm		75mm		75mm		up to 0.6m	
Antenna		-		-		-		-		yes		yes		yes		yes		yes	
FIFO depth [byte]		64		64		64		64		n.a.		n.a.		n.a.		n.a.		n.a.	
Host interface		8-bit parallel		8-bit parallel, SPI		8-bit parallel, SPI		8-bit parallel, SPI		USB		USB/R232		USB		USB		RS232	
RF Interface		fully integrated		fully integrated		fully integrated		fully integrated		MF RC500		MF RC500		SL RC400		discrete		-	
Analog interface		13.56		13.56		13.56		13.56		13.56		13.56		13.56		13.56		13.56	
Carrier frequency [MHz]		100% ASK		100% ASK		10% & 100% ASK		10% & 100% ASK		100% ASK		100% ASK		100% ASK		10% & 100% ASK		10% & 100% ASK	
Modulation		106		106/212/424/848		106/212/424/848		106/212/424/848		106		106		106		106		-	
Baudrate ISO 14443 [kBaud]		-		-		-		-		-		-		-		-		-	
Baudrate ISO 15693 [kBaud]		-		-		-		-		-		-		-		-		-	
Standards & Protocols		ISO 14443 A		yes		yes		yes		yes		yes		yes		-		-	
ISO 14443 B		-		-		-		-		-		-		-		-		-	
ISO 15693		-		-		-		-		-		-		-		-		yes	
mifare® Classic protocol		yes		yes		yes		yes		yes		yes		yes		-		-	
I*CODE 1 protocol		-		-		-		-		-		-		-		-		yes	
Security Features		mifare® Classic		yes		yes		yes		yes		yes		yes		-		-	
Exception Sensors		yes		yes		yes		yes		yes		yes		yes		-		-	
Additional Product Information		Supply voltage digital [V]		5		3.3 or 5		3.3 or 5		5		5		5		5		24	
Supply voltage analog [V]		5		5		5		5		5		5		5		5		5	
Power down mode current typ. [µA]		2		2		2		2		2		2		2		2		2	
Wake up time [µs]		1000		1000		1000		1000		1000		1000		1000		1000		1000	
Temperature range [°C]		-25 ... +85		-25 ... +85		-25 ... +85		-25 ... +85		-25 ... +85		-25 ... +85		-25 ... +85		-25 ... +85		-25 ... +85	
Package		SO32		SO32		SO32		SO32		SO32		SO32		SO32		SO32		SO32	
Approvals		n.a.		n.a.		n.a.		n.a.		n.a.		n.a.		n.a.		n.a.		n.a.	
EMC		yes		yes		yes		yes		yes		yes		yes		yes		yes	
Software Support		yes		yes		yes		yes		yes		yes		yes		yes		yes	

ISO 7816

Philips' Contact Smart Card Reader ICs

	Analog Interface					Analog & UART	Analog & UART & µC				
Product Features	OM5926	TDA8002C	TDA8004A	TDA8020	TDA8024	TDA8007B	TDA8006	TDA8008	TDA8029	TDA8030	TDA8031
Analog interfaces	1 SIM/VIM card	1 card	1 card	2 cards	1 card	2 cards	1 card	2 cards	1 card	1 card	1 card
ISO 7816 UART	-	-	-	-	-	-	yes	yes	yes	yes	yes
ISO 7816 dedicated timers	-	-	-	-	-	yes	-	yes	yes	yes	yes
µC-core	-	-	-	-	-	-	80C52	80C51RB+	80C51RB+	80C51RB+	80C51RB+
ROM/OTP [kbyte]	-	-	-	-	-	-	16 ROM	16 OTP	16 ROM	16 OTP	16 ROM
RAM [byte]	-	-	-	-	-	-	256 + 1k	768	768	768	768
Host interface	I/C	I/O lines	I/O lines	I/C	I/O lines	8-bit parallel	serial	serial	serial	USB	USB
Card Interface											
ESD protection on ISO pads [kV]	6	6	6	6	6	6	6	6	6	6	6
Auxiliary protected I/O lines	-	2	2	2	2	2x2	2	2 x 2	-	2	2
Vcc-card power supply [V]	3 & 5	3 & 5	3 & 5	3 & 5	3 & 5	1.8 & 3 & 5	3 & 5	1.8 & 3 & 5	1.8 & 3 & 5	1.8 & 3 & 5	1.8 & 3 & 5
Card supply current @ 5V Vcc [mA]	40	55	65	2 x 55	80	2 x 55	65	2 x 55	65	55	55
Card supply current @ 3V Vcc [mA]	40	55	65	2 x 50	65	2 x 50	50	2 x 50	50	50	50
Card supply current @ 1.8V Vcc [mA]	-	-	-	-	-	30	30	30	30	30	30
Card clock frequency max. [MHz]	10	12	20	20	20	20	10	20	20	12	12
Card activation time max. [µs]	150	220	220	135	220	130	220	130	225	250	250
Card deactivation time max. [µs]	100	90	90	110	100	150	100	100	100	100	100
Protocol Support											
Synchronous card management	-	yes	-	-	-	yes	yes	yes	yes	yes	yes
Asynchronous protocol T=0 and T=1	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Security Features											
Voltage supervisor & over current detection	yes & no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Current protection on VCC, I/O, RST, CLK	-	-	-	yes	yes	yes	-	yes	yes	yes	yes
Additional Product Information											
Power supply [V]	2.5 to 6	3 to 6.5	2.7 to 6.5	2.5 to 6.5	2.7 to 6.5	2.7 to 6	4.2 to 6	2.7 to 6	2.7 to 6	4.2 to 5.5	4.2 to 5.5
Power down mode current max. [µA]	2	70	-	150	-	350	250	100	20 (shut down)	500 (suspend)	500 (suspend)
Temperature range [°C]	-40 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-40 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85	-25 ... +85
Package	HVQFN20	SO28/LQFP32	SO28	LQFP32	SO28/LQFP32	LQFP48	QFP44/QFP64	LQFP80	LQFP32	LQFP64	LQFP64
Software Support (EMV)	-	-	-	-	-	yes	yes	yes	yes	yes	yes
EMV compliance	-	EMV3.1.1 / EMV4.0	EMV3.1.1 / EMV4.0	EMV3.1.1 / EMV4.0	EMV3.1.1 / EMV4.0	EMV3.1.1 / EMV4.0	EMV3.1.1	EMV3.1	EMV3.1.1 / EMV4.0	EMV3.1.1	EMV3.1.1

